



Operations Research

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Solution to the Training Exercises, Sheet 2

Training Exercises

Exercise T2.1. *Simplex algorithm*

Problem

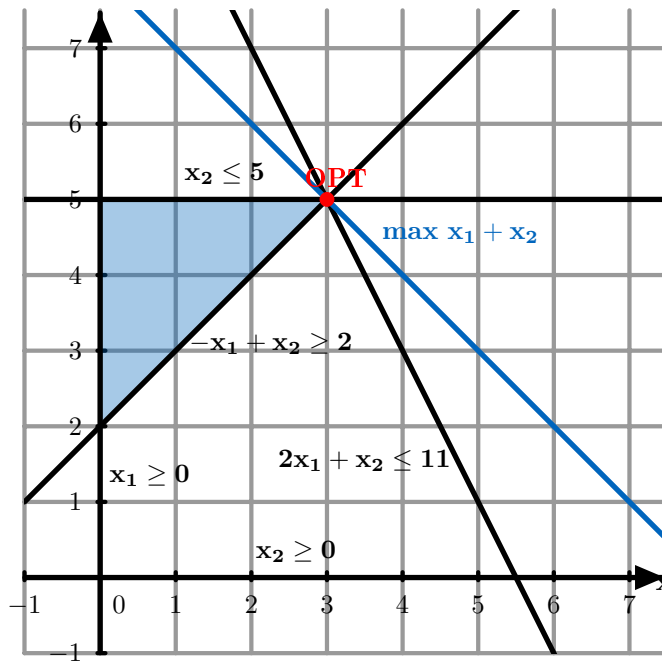
Consider the following linear program (LP):

$$\begin{array}{ll}\text{Max} & x_1 + x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \leq 11 \\ & x_2 \leq 5 \\ & x \in \mathbb{R}_{\geq 0}^2\end{array}$$

- Solve the LP graphically.
- Convert the LP into standard form.
- Is the LP in canonical form? Justify.
- Solve the LP with the simplex algorithm in tableau notation using the Big M method.
- Is the LP degenerate? Justify your answer. How can you read such degeneracy from the optimal tableau of the simplex algorithm?
- The final tableau of task part (d) is an optimal simplex tableau. Does another optimal simplex tableau exist? Give one or justify why no other optimal simplex tableau can exist.

Solution

- Graphical solution:



b)

$$\begin{aligned}
 &\text{Max} && x_1 + x_2 \\
 &\text{s.t.} && -x_1 + x_2 - x_3 = 2 \\
 &&& 2x_1 + x_2 + x_4 = 11 \\
 &&& x_2 + x_5 = 5 \\
 &&& x \in \mathbb{R}_{\geq 0}^5
 \end{aligned}$$

c) The LP is not in canonical form because there is no basic variable for the first constraint.

d) We need an additional artificial variable $w_1 \geq 0$ for the first constraint.

$$\begin{aligned}
 &\text{Max} && x_1 + x_2 - Mw_1 \\
 &\text{s.t.} && -x_1 + x_2 - x_3 + w_1 = 2 \\
 &&& 2x_1 + x_2 + x_4 = 11 \\
 &&& x_2 + x_5 = 5 \\
 &&& x \in \mathbb{R}_{\geq 0}^5 \\
 &&& w \in \mathbb{R}_{\geq 0}
 \end{aligned}$$

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	w_1	
z	-1	-1				M	0
w_1	-1	1	-1			1	2
x_4	2	1		1			11
x_5		1			1		5

The tableau must first be converted into canonical form. To do this, the variable w_1 in the objective function is eliminated.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	w_1	
z	$-1 + M$	$-1 - M$	M				$-2M$
w_1	-1	1	-1			1	2
x_4	2	1		1			11
x_5		1			1		5

Now phase 2 of the simplex algorithm is executed. Since the artificial slack variable

w_1 becomes a non-base variable by the base swap $w_1 \leftrightarrow x_2$, the w_1 column need not be considered from now on. In the following, it is only included for the sake of completeness.

Base swap $w_1 \leftrightarrow x_2$

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	w_1	
z	-2		-1			$1 + M$	2
x_2	-1	1	-1			1	2
x_4	3		1	1		-1	9
x_5	1		1		1	-1	3

Base swap $x_5 \leftrightarrow x_1$
(also possible: $x_4 \leftrightarrow x_1$)

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	w_1	
z			1		2	$-1 + M$	8
x_2		1			1		5
x_4			-2	1	-3	2	0
x_1	1		1		1	-1	3

The stopping rule is satisfied (only nonnegative entries in the z-line). The optimal solution is given by $x^* = (3, 5)$ with associated objective function value $z_{\max} = 8$.

Alternative base swap $x_4 \leftrightarrow x_1$: Theoretically, the base swap $x_4 \leftrightarrow x_1$ would also have been possible. However, after this base swap the tableau is not optimal, so another base swap $x_5 \leftrightarrow x_3$ would be necessary.

Base swap $x_4 \leftrightarrow x_1$

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	w_1	
z			$-1/3$	$2/3$		$1/3 + M$	8
x_2		1	$-2/3$	$1/3$		$2/3$	5
x_5			$2/3$	$-1/3$	1	$-2/3$	0
x_1	1		$1/3$	$1/3$		$-1/3$	3

Base swap $x_5 \leftrightarrow x_3$

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	w_1	
z				$1/2$	$1/2$	M	8
x_2		1			1		5
x_3			1	$-1/2$	$3/2$	-1	0
x_1	1			$1/2$	$-1/2$		3

The stopping rule is satisfied (only nonnegative entries in the z-line). The optimal solution is given by $x^* = (3, 5)$ with associated objective function value $z_{\max} = 8$.

- e) Yes, the LP is degenerate. More than two constraints intersect at the optimal point x^* , see graph from task part (a). In the simple tableau, such a degeneracy can be seen if at least one basis variable takes the value 0. For example, this is the case for the base variable x_4 after the base swap $x_5 \leftrightarrow x_1$.
- f) Since in the optimal simplex tableau from task part (d) one base variable takes the value 0, one can perform another pivot step without changing the objective function value. For example, given the optimal tableau resulting from the base swap $x_5 \leftrightarrow x_1$,

one can perform the base swap $x_4 \leftrightarrow x_3$. This produces the final tableau which resulted in task part (d) from the alternative base swap $x_4 \leftrightarrow x_1$.

Base swap $x_4 \leftrightarrow x_3$

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	w_1	
z				$1/2$	$1/2$	M	8
x_2		1			1		5
x_3			1	$-1/2$	$3/2$	-1	0
x_1	1			$1/2$	$-1/2$		3

Exercise T2.2. Simplex algorithm

Problem

- a) The following linear program is given in tableau notation. Specify all pivot elements by means of which the objective function value can be increased in the next iteration of the simplex algorithm. For each of these pivot elements, also name the entering and exiting basis variable. Justify your answer.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	x_6	
z	-1	-2	1	0	0	0	0
x_4	-1	2	1	1	0	0	3
x_5	2	4	4	0	1	0	4
x_6	-1	0	2	0	0	1	6

- b) The following linear program is given in tableau notation. Here x_1 and x_2 are structure variables, whereas x_3 , x_4 and x_5 are slack variables. Decide whether the underlying linear program is optimally solvable, intractable, or unbounded. If optimal solutions exist, explicitly state all optimal solutions $x^* = (x_1, x_2)$ for the underlying linear program. Justify your answer.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	0	0	3	1	0	1
x_2	0	1	1	1	0	1
x_1	1	0	-1	1	0	1
x_5	0	0	-1	-1	1	0

- c) The following linear program is given in tableau notation. Here x_1 and x_2 are structure variables, whereas x_3 , x_4 and x_5 are slack variables. Decide whether the underlying linear program is optimally solvable, intractable, or unbounded. If optimal solutions exist, explicitly state all optimal solutions $x^* = (x_1, x_2)$ for the underlying linear program. Justify your answer.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	0	0	0	1	0	5
x_3	0	0	1	-2	3	3
x_2	0	1	0	1	-1	2
x_1	1	0	0	0	1	3

- d) The following linear program is given in tableau notation. Here x_1 and x_2 are structure variables, whereas x_3 and x_4 are slack variables. The variable w_1 is an artificial variable, which was introduced to be able to solve the linear program using the Big M method. Decide whether the underlying linear program is optimally solvable, intractable, or unbounded. If optimal solutions exist, explicitly state all optimal solutions $x^* = (x_1, x_2)$ for the underlying linear program. Justify your answer.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	M	0	$1 + M$	M	0	$1 - M$
x_2	1	1	1	0	0	1
w_1	-1	0	-1	-1	1	1

Solution

- a) $\mathbf{x}_5 \leftrightarrow \mathbf{x}_1$: The objective function coefficient of x_1 is negative. Moreover, the entry corresponding to the x_5 column is positive in the x_1 column. Since $x_5 > 0$, the objective function value can be increased.
- $\mathbf{x}_5 \leftrightarrow \mathbf{x}_2$: The objective function coefficient of x_2 is negative. Since $3/2 > 4/4$, x_5 must exit the basis. (So the base swap $x_4 \leftrightarrow x_2$ is *not* possible! In this base swap, the base variable x_5 would become negative). Since $x_5 > 0$, the objective function value can be increased.
- The pivot elements are the intersections of the pivot row and column, respectively.
- b) Since the objective function row contains only nonnegative coefficients, the simplex tableau is optimal. There exists exactly one optimal solution for the underlying linear program, which can be read directly from the tableau: $x^* = (1, 1)$. Since the base variable x_5 has the value zero, the base solution is degenerate. Thus, multiple optimal simplex tableaux may exist under some circumstances. However, for all of them, the optimal solution $x^* = (x_1, x_2)$ for the underlying linear program would be the same.
- c) The optimal solution $x_1^* = (3, 2)$ can be read directly from the optimal tableau. Since the objective function coefficient of the non-base variable x_5 is zero, one can perform the base swap $x_3 \leftrightarrow x_5$ without changing the objective function value. Thus, another solution of the linear program would be $x_2^* = (2, 3)$. Moreover, due to the convexity of linear programs, all points on the line between $(3, 2)$ and $(2, 3)$ are optimal. These are given by $(1 - \alpha) \cdot (3, 2) + \alpha \cdot (2, 3) = (3 - \alpha, 2 + \alpha)$ with $\alpha \in [0, 1]$.
- d) The given tableau cannot be optimized further because all objective function coefficients are nonnegative. However, the artificial variable w_1 is still part of the basis. Thus, the underlying LP is intractable.

Self-study Exercises

Exercise S2.1. Simplex algorithm

Problem

- a) The following linear program is given in tableau notation. In what value ranges must the parameters α and β lie so that in the next iteration of the simplex algorithm the objective function value can be increased by x_1 entering the basis and x_2 leaving the basis? Justify your answer.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	x_6	
z	α	0	2	0	3	0	0
x_4	-1	0	-2	1	5	0	1
x_6	2	0	0	0	1	1	4
x_2	6	1	7	0	3	0	β

- b) The following linear program is given in tableau notation. Give the largest possible value ranges for the parameters α and β so that one can read directly from the simplex tableau that the underlying linear program is unbounded. Justify your answer.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	x_6	
z	0	3	0	α	1	0	0
x_3	0	-1	1	-1	3	0	1
x_6	0	-4	0	-3	1	1	3
x_1	1	2	0	β	1	0	2

- c) The following linear program is given in tableau notation. Give a value for each of the parameters α and β such that degenerate basis solutions exist. Justify your answer.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	x_6	
z	-1	-4	-1	0	0	0	0
x_4	-1	2	β	1	0	0	4
x_5	2	1	2	0	1	0	9
x_6	α	6	3	0	0	1	12

Solution

- a) $\alpha < 0$ and $0 < \beta \leq 12$

Let x_α denote the non-base variable associated with α and x_β denote the base variable associated with β . For the objective function value to be increased and x_α to be selected as the entering basis variable, $\alpha < 0$ must hold. For x_β to be selected as the exiting basis variable, $\beta/6 \leq 4/2$ must hold. From this follows $\beta \leq 12$. For the objective function value to be increased, $\beta > 0$ must also hold, of course.

b) $\alpha < 0$ and $\beta \leq 0$

Let x_α denote the non-base variable associated with α . For the objective function value to be increased and x_α to be selected as the entering basis variable, $\alpha < 0$ must hold. In addition, in order to increase x_α infinitely, $\beta \leq 0$ must hold.

c) $\alpha = -3$ and $\beta = 1$

Let r_α and r_β denote the restrictions associated with α and β , respectively. It can be seen that these two restrictions are multiples of each other when $\alpha = -3$ and $\beta = 1$. Thus the two restrictions lie exactly on top of each other, degenerating the optimal solution.

In addition, there are other parameter assignments of α and β that would also lead to degenerate basis solutions.

Exercise S2.2. Big M method

Problem

Consider the following linear program (LP):

$$\begin{array}{ll} \text{Max} & x_1 + x_2 \\ \text{s.t.} & x_1 + 2x_2 = 4 \\ & x_1 - x_2 = 1 \\ & 2x_1 + x_2 \geq 5 \\ & x \in \mathbb{R}_{\geq 0}^2 \end{array}$$

Compute an optimal basis solution for the LP with the simplex algorithm using the Big M method.

Solution

We first transform the LP into the standard form.

$$\begin{array}{ll} \text{Max} & x_1 + x_2 \\ \text{s.t.} & x_1 + 2x_2 = 4 \\ & x_1 - x_2 = 1 \\ & 2x_1 + x_2 - x_3 = 5 \\ & x \in \mathbb{R}_{\geq 0}^3 \end{array}$$

The LP is not in canonical form since there is no basic variable for any of the three constraints. We therefore need three additional artificial variables $w_1, w_2, w_3 \in \mathbb{R}_{\geq 0}$.

$$\begin{array}{ll} \text{Max} & x_1 + x_2 - Mw_1 - Mw_2 - Mw_3 \\ \text{s.t.} & x_1 + 2x_2 + w_1 = 4 \\ & x_1 - x_2 + w_2 = 1 \\ & 2x_1 + x_2 - x_3 + w_3 = 5 \\ & x \in \mathbb{R}_{\geq 0}^3 \\ & w \in \mathbb{R}_{\geq 0}^3 \end{array}$$

Base	Coefficients						Res.
	x_1	x_2	x_3	w_1	w_2	w_3	
z	-1	-1		M	M	M	0
w_1	1	2		1			4
w_2	1	-1			1		1
w_3	2	1	-1			1	5

The tableau must first be converted into canonical form. To do this, the variables w_1 , w_2 and w_3 are eliminated in the objective function.

Base	Coefficients						Res.
	x_1	x_2	x_3	w_1	w_2	w_3	
z	$-1 - 4M$	$-1 - 2M$	M				$-10M$
w_1	1	2		1			4
w_2	1	-1			1		1
w_3	2	1	-1			1	5

Now phase 2 of the simplex algorithm is executed.

Base swap $w_2 \leftrightarrow x_1$

Base	Coefficients						Res.
	x_1	x_2	x_3	w_1	w_2	w_3	
z		$-2 - 6M$	M		$1 + 4M$		$1 - 6M$
w_1		3		1	-1		3
x_1	1	-1			1		1
w_3		3	-1		-2	1	3

Base swap $w_1 \leftrightarrow x_2$
(also possible: $w_3 \leftrightarrow x_2$)

Base	Coefficients						Res.
	x_1	x_2	x_3	w_1	w_2	w_3	
z			M	$2/3 + 2M$	$1/3 + 2M$		3
x_2		1		$1/3$	$-1/3$		1
x_1	1			$1/3$	$2/3$		2
w_3			-1	-1	-1	1	0

The simplex algorithm terminates because the z -row contains only nonnegative entries. However, the artificial variable w_3 is still part of the basis. Yet, since the right-hand side of the associated equation is zero, an additional base swap $w_3 \leftrightarrow x_3$ can be performed without changing the objective function value.

Base swap $w_3 \leftrightarrow x_3$

Base	Coefficients						Res.
	x_1	x_2	x_3	w_1	w_2	w_3	
z				$2/3 + M$	$1/3 + M$	M	3
x_2		1		$1/3$	$-1/3$		1
x_1	1			$1/3$	$2/3$		2
x_3			1	1	1	-1	0

The optimal solution is given by $x^* = (2, 1)$ with associated objective function value $z_{\max} = 3$.

Exercise S2.3. Simplex tableau special cases

Problem

Consider the following linear programs in tableau representation:

a)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z				0	1	10
x_3			1	2	-1	5
x_2		1		1	0	4
x_1	1			-2	1	2

b)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-50	-45				0
x_3	6	5	1			60
x_4	6	5		1		0
x_5	10	20			1	150

c)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	2	-1	0			2
x_3	-2	-1	1			2
x_4	1	0		1		4
x_5	0	-2			1	1

e)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-7	-5				0
x_3	3	4	1			24
x_4	2	1		1		10
x_5	-1				1	-1

d)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-50	-45				0
x_3	6	5	1			60
x_4	16	25		1		320
x_5	10	20			1	150

f)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-7	-5				0
x_3	3	4	1			24
x_4	2	1		1		10
x_5	1				1	-1

Indicate in each case which special case it is and how to recognize this special case in the tableau representation.

Furthermore, indicate how the respective special case affects the flow of the simplex algorithm.

Solution

- a) **Multiple optimal solutions:** This can be seen from the fact that in the optimal tableau, the objective function coefficient of at least one non-base variable takes the value of zero.

Effect on the simplex: The simplex terminates here at the first optimal solution found. However, further optimal solutions can be generated by appropriate simplex steps. Note that any convex combination of two optimal solutions is also an optimal solution. The current solution is $x_1^* = (2, 4, 5, 0, 0)$. By base swapping $x_3 \leftrightarrow x_4$ we obtain a new tableau with $x_2^* = (7, 1.5, 0, 2.5, 0)$. All optimal solutions can thus be represented by $x_\alpha^* = \alpha \cdot x_1^* + (1 - \alpha) \cdot x_2^*$ with $\alpha \in [0, 1]$.

- b) **Redundant constraints:** This means that a linear combination of other constraints has the same left side, but a smaller right side. In the simplex tableau, such redundancy can be demonstrated by the existence of such a linear combination. A particularly simple form of redundancy can also be recognized by the fact that in the course of the simplex two or more tableau rows appear whose coefficients are identical in the non-base variable area (here: constraints 1 and 2).

Effect on the simplex: None special. If such a redundancy is discovered at the beginning, the simpler problem can be solved by the simplex and the same optimal solution is obtained as in the original problem.

Degenerate corner: This special case arises when more than dimension many constraints describe the same corner. Therefore, when there is degeneracy, there are always redundant constraints. In the simplex tableau, degeneration can be recognized by the fact that at least one base variable takes the value zero (here: x_4).

Effect on the simplex: A base swap is performed without improving the objective function value (risk that the primary simplex will not leave the degenerate corner and

will never terminate).

- c) **Primary unrestrictedness** resp. **dual infeasibility**: The admissible solution space is unbounded and each admissible solution can be further improved. This can be seen by the fact that in a possible pivot column there are only entries with values ≤ 0 . In our example, the variable x_2 can be increased to infinity, thereby also increasing the objective function value arbitrarily.

Effect on the simplex: The variable x_2 cannot be swapped into the base, since we cannot determine a pivot row or a pivot element. Since we can no longer perform the pivot step, the simplex algorithm terminates with $OPT = +/\infty$ (1 of 3 possible outputs of the simplex).

- d) **Redundant constraint**: This means that a linear combination of other constraints has the same left side, but a smaller right side. Here, constraint 2 is redundant because the linear combination (constraint1 + constraint3) forms a more restrictive constraint.

Effect on the simplex: see above.

- e) **No canonical form**: Here there is no admissible initial solution. This can be recognized by the fact that one cannot produce a canonical form in the initial tableau without negative entries appearing in the right-hand side.

Effect on the simplex: Since we now have an initial solution, but it is not admissible because of the negative right-hand side, we must first find an admissible initial solution. So we have to use simplex phase I, the big M method, or the dual simplex.

- f) **No canonical form + Primal infeasibility** resp. **dual unconstraintness**: No admissible solution exists here. This can be seen in the 3rd constraint. Since all variables must be ≥ 0 , there is no occupancy such that $a_3^t x < 0$.

Effect on the simplex: If one sees this immediately, there is no need to run the simplex at all. If one does not see the impropriety immediately, one would have to use the big M method (or phase 1 of the simplex algorithm). However, in the course of the 2nd phase of the simplex algorithm, an artificial slack variable would then always be a base variable, which means that $Z_P = \emptyset$.

Want more info? Read chapter 2.5.2 "Sonderfälle von LPs und ihre Identifikation" in Domschke-Drexler.

Exercise S2.4. Simplex tableau general and special cases

Problem

- a) Transform the following LP into canonical form.

$$\begin{array}{ll} \text{Max} & x_1 + 2x_2 \\ \text{s.t.} & -2x_1 + x_2 \geq -2 \\ & -x_1 + x_2 \leq 3 \\ & x_1 + x_2 \leq 7 \\ & x \in \mathbb{R}_{\geq 0}^2 \end{array}$$

b) Perform one iteration of the simplex algorithm. State the incoming and outgoing basic variables.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-3			2		6
x_3	1		1	1		5
x_2	-1	1		1		3
x_5	2			-1	1	4

Is the resulting tableau optimal?

c) For the following LP in tableau notation, provide the largest possible ranges for α and β such that one can read directly from the simplex tableau that the underlying LP is unbounded. Justify your answer.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	α			2		6
x_3	-1		1	1		5
x_2	-1	1		1		3
x_5	β			1	-1	-4

d) Can the same vertex be visited more than once during the simplex algorithm? Justify your answer.

e) Can a vertex be revisited after a different vertex was visited? Justify your answer.

Solution

a)

$$\begin{aligned}
 \text{Max}_{\text{s.t.}} \quad & x_1 + 2x_2 = z \\
 & 2x_1 - x_2 + x_3 = 2 \\
 & -x_1 + x_2 + x_4 = 3 \\
 & x_1 + x_2 + x_5 = 7 \\
 & x \in \mathbb{R}_{\geq 0}^5
 \end{aligned}$$

b) Basis swap $x_5 \leftrightarrow x_1$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z				1/2	3/2	12
x_3			1	3/2	-1/2	3
x_2		1		1/2	1/2	5
x_1	1			-1/2	1/2	2

Stopping criterium met, only positive entries in the objective row.

c) $\alpha < 0$: Implies that x_1 is the incoming basic variable, $\beta \geq 0$: Note that the tableau is not in canonical form. The last row has to be multiplied by -1 . If all coefficients are negative in the x_1 columns, then the LP is unbounded. Hence, $-\beta \leq 0 \Leftrightarrow \beta \geq 0$.

d) Yes, if the problem is degenerate.

e) No, since the objective strictly increases when visiting a different vertex.